

Kakarla Venkata Rama Sai Pavan Shanmukh

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Career Objective

Adaptable B.Tech student and full-stack developer specializing in scalable distributed ledger analytics and secure real-time systems to drive data-driven architectural innovation.

Education

KL University 9.51 CGPA	Aug 2023 - Present
Bachelor of Technology (B.Tech) Computer Science and Information Technology	Vijayawada A.P
Narayana Educational Institutions 9.31 CGPA	Aug 2021 - Jul 2023
Intermediate Andhra Pradesh Board of Intermediate Education	Vijayawada
Nalanda Vidya Niketan 8.18 CGPA	Jul 2020 - May 2021
10th Grade Central Board of Secondary Education	Vijayawada

Experience

MeshaSec Technologies Pvt Ltd.	May 2026 - Jul 2026
Cybersecurity Research & AI Intern	
Gained hands-on experience in web application security and backend development by performing vulnerability assessments using DAST tools, solving 10+ security lab machines, and developing/testing REST APIs using Node.js, Express.js, and Postman.	
Cisco Networking Academy	May 2024 - Jul 2024
Virtual Intern	
Hands-on networking with Cisco Packet Tracer. Designed and simulated network topologies; implemented and tested routing algorithms for optimized performance.	

Skills

Core: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Linux, OOPS Through Java, Git and GitHub
Languages: Java, Python, HTML5, Linux, CSS3, JavaScript
Tools, Frameworks and Databases: React.js, Spring Boot, Node.js, Power BI, CI/CD pipelines, MySQL, PostgreSQL

Projects

SelectShans: Ransomware Pre-Encryption Detection System	Feb 2026 - Sep 2026
Abstract: A real-time Watchdog-based behavioral monitoring and threshold logic to identify ransomware encryption with 80% accuracy.	
Technologies, Tools & Software: Python, Watchdog, Flask, MySQL, REST APIs, and behavioral anomaly monitoring.	
Outcome and Social Relevance: Minimized detection latency to 10 seconds & safeguarding critical infrastructure against data loss.	
Technologies: Python, React.js, Network Security, JavaScript, Git, GitHub	
OmniBank: Full-Stack Financial Management System	Oct 2025 - Mar 2026
Abstract: A secure full-stack banking platform featuring real-time transactions, MFA, and Automated CI/CD pipelines.	
Technologies, Tools & Software: React.js, Vite, Spring Boot, Spring Security, MySQL, REST APIs, and CI/CD tools.	
Outcome and Social Relevance: Provided a platform for transparent financial tracking.	
Technologies: JavaScript, CI / CD, Docker, Kubernetes, Jenkins	
LexiTree: Optimized Text Analysis via Self-Balancing AVL Trees	Nov 2023 - Feb 2024
Abstract: Decreased AVL Tree's performance to O(log n) word frequency analysis & visualization for large-scale textual datasets.	
Technologies, Tools & Software: AVL Tree, UI Frameworks, and Performance Analysis modules.	
Outcome and Social Relevance: Optimized large-scale data processing.	
Technologies: React.js, Web Technologies, Data Structures and Algorithms, Git, GitHub	

Certifications and Achievements

Microsoft Certified - Azure Fundamentals	Mar 2026
Acquired basic knowledge of cloud services and how they are provided via Microsoft Azure.	
Optimized Antenna Design for WBAN Applications	Dec 2024 - Present
A high-efficiency, flexible design featuring a metamaterial-backed ground plane to ensure stable on-body communication.	

AWS Certified - Developer Associate
Learned to develop, deploy, secure, and optimize cloud-native applications on AWS using services like Lambda, EC2, S3, DynamoDB, API Gateway, IAM, CloudWatch, and CI/CD tools while following AWS best practices for scalability, reliability, and security.

Coding Profiles

Leetcode: [PavanShanmukhKakarla](#) Codechef: [kakarla_pavan1](#) Codeforces: [PavanShanmukhKakarla](#)